

CS 210 Problem Set 1

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1 Problem 1.2

Implement a merge sort that avoids defining a sentinel value.

Solution. We can implement the merging procedure to array C without sentinels by having iteration variables the two arrays A and B and taking note their position. This is given by the algorithm below:

```
function MERGE(A, B)
   $C \leftarrow []$ 
   $i \leftarrow 0$ 
   $j \leftarrow 0$ 
  repeat
    if  $A[i] \leq B[j]$  then
       $\text{append}(C, A[i])$ 
      if  $i + 1 < \text{length}(A)$  then
         $i \leftarrow i + 1$ 
      end if
    else
       $\text{append}(C, B[j])$ 
      if  $j + 1 < \text{length}(B)$  then
         $j \leftarrow j + 1$ 
      end if
    end if
  until  $i + 1 = \text{length}(A)$  &  $j + 1 = \text{length}(B)$ 
end function
```

Example. Consider merging two arrays:

i	↓				
A	1	2	4	5	
j	↓				
B	2	3	6	8	9
C					

i		↓			
A	1	2	4	5	
j	↓				
B	2	3	6	8	9
C	1				

i			↓		
A	1	2	4	5	
j	↓				
B	2	3	6	8	9
C	1	2			

i			↓		
A	1	2	4	5	
j		↓			
B	2	3	6	8	9
C	1	2	2		

i			↓		
A	1	2	4	5	
j			↓		
B	2	3	6	8	9
C	1	2	2	3	

i				↓	
A	1	2	4	5	
j			↓		
B	2	3	6	8	9
C	1	2	2	3	4

i				↓		
A	1	2	4	5		
j				↓		
B	2	3	6	8	9	
C	1	2	2	3	4	5
						6

i				↓			
A	1	2	4	5			
j				↓			
B	2	3	6	8	9		
C	1	2	2	3	4	5	6
						8	9

Since i and j are both pointing to end of A and B , the procedure will stop.

2 Problem 1.3

Implement a merge sort that divides the array into three equal parts, sorts them, and does a three-way merge. Empirically compare its running time with standard merge sort.

Solution. The three-way merge sort is given by the following algorithm:

```

function MERGESORT3(A)
   $n \leftarrow \text{length}(A)$ 
  if  $n \leq 1$  then return A
  end if
   $x \leftarrow \lceil n/3 \rceil$ 
   $y \leftarrow \lceil 2n/3 \rceil$ 
  return MERGE3( $A[0 \dots x - 1]$ ,  $A[x \dots y - 1]$ ,  $A[y \dots n - 1]$ )
end function

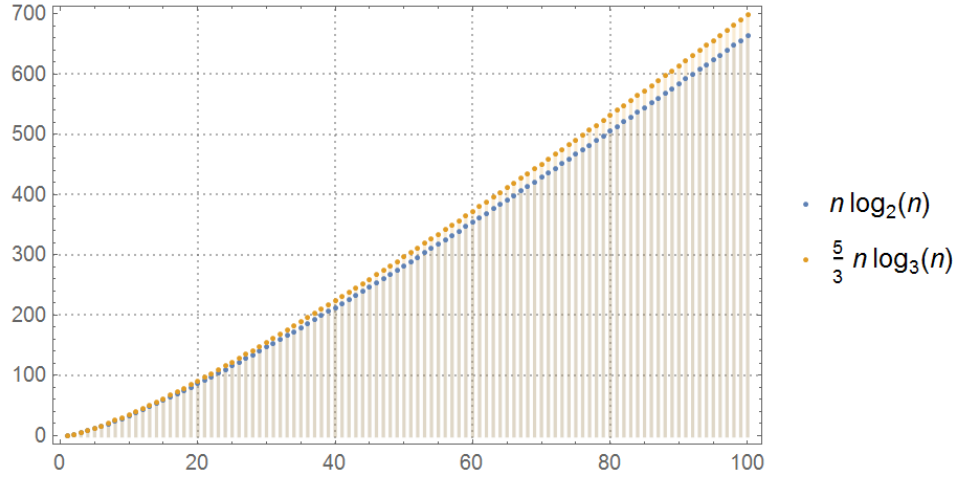
function MERGE3( $A_0, A_1, A_2$ )
  return MERGE( $(A_0, \text{Merge}(A_1, A_2))$ )
end function

```

Comparing the running time with standard merge sort, we have

N	2-way	3-way
1	0	0
2	2	2.1031
3	4.75489	5
4	8	8.4124
5	11.6096	12.2081
6	15.5098	16.3093
7	19.6515	20.6645
8	24	25.2372
9	28.5293	30
10	33.2193	34.9317

Comparing asymptotically as N goes large, we have the following graph:



The 3-way merge sort generally takes more time than the 2-way merge sort.

3 Problem 1.6

Analyze the number of compares used by the three-way merge sort.

Solution. The number of compares on the inner merging is $\frac{n}{3} + \frac{n}{3}$ and $\frac{n}{3} + \frac{2n}{3}$ for the outer merging. Then the total number of compares for the whole merging is $\frac{n+n+n+2n}{3} = \frac{5n}{3}$. Given the three-way merge algorithm above, the total number of compares is given by

$$C_n = 3C_{\lfloor n/3 \rfloor} + \frac{5n}{3}, \quad C_1 = 0.$$

Solving the recurrence relation in terms of $n = 3^m \geq 1$, we have:

$$\begin{aligned} C_{3^m} &= 3C_{3^{m-1}} + 5 \cdot 3^{m-1} \\ \frac{C_{3^m}}{3^m} &= \frac{C_{3^{m-1}}}{3^{m-1}} + \frac{5}{3} \\ \frac{C_{3^m}}{3^m} &= \frac{C_{3^{m-2}}}{3^{m-2}} + \frac{5}{3} + \frac{5}{3} = \frac{C_{3^0}}{3^0} + \frac{5m}{3} = \frac{5m}{3} \\ C_{3^m} &= \frac{5}{3}m \cdot 3^m \\ \boxed{C_n} &= \frac{5}{3}n \log_3(n) \end{aligned}$$

4 Problem 2.8

For f linear, express the solution to the recurrence

$$a_n = f(a_{n-1}, a_{n-2})$$

for $n > 1$ in terms of $a_0, a_1, f(0, 0)$.

Solution. The solution is

$$a_n = a_0 u_n + a_1 v_n + C * f(0, 0)$$

where

$$u_n = f(u_{n-1}, u_{n-2})$$

for $n > 1$ with $u_0 = 1$ and $u_1 = 0$. and

$$v_n = f(v_{n-1}, v_{n-2})$$

for $n > 1$ with $v_0 = 0$ and $v_1 = 1$ where C is a constant obtained from the method of undetermined coefficients.

Follow-up. Find the solutions to

$$a_n = f(a_{n-1}, a_{n-2}) - f(0, 0)$$

for $a_1 = 1, a_0 = 0$ and $a_0 = 1, a_1 = 0$.

Solution. This is a homogeneous linear recurrence.

Case 1: $a_1 = 1, a_0 = 0$. The solution is equal to

$$v_n = f(v_{n-1}, v_{n-2})$$

where $v_0 = 0$ and $v_1 = 1$.

Case 2: $a_1 = 0, a_0 = 1$. The solution is equal to

$$u_n = f(u_{n-1}, u_{n-2})$$

where $v_0 = 1$ and $v_1 = 0$.

5 Problem 2.14

Write down the solution to

$$a_n = x_n a_{n-1} + y_n$$

for $n > t$ in terms of the x 's, y 's and the initial value a_t .

Solution. Note that the summation factor is $x_n x_{n-1} x_{n-2} \dots x_t$. Divide both sides by summation factor and telescope until we get the closed form.

$$\begin{aligned}
\frac{a_n}{x_n x_{n-1} \dots x_t} &= \frac{a_{n-1}}{x_{n-1} x_{n-2} \dots x_t} + \frac{y_n}{x_n x_{n-1} \dots x_t} \\
&= \frac{a_{n-2}}{x_{n-2} x_{n-3} \dots x_t} + \frac{y_{n-1}}{x_{n-1} x_{n-2} \dots x_t} + \frac{y_n}{x_n x_{n-1} \dots x_t} \\
&= \frac{a_{n-3}}{x_{n-3} x_{n-4} \dots x_t} + \frac{y_{n-2}}{x_{n-2} x_{n-3} \dots x_t} + \frac{y_{n-1}}{x_{n-1} x_{n-2} \dots x_t} + \frac{y_n}{x_n x_{n-1} \dots x_t} \\
\frac{a_n}{x_n x_{n-1} \dots x_t} &= \frac{a_t}{x_t} + \frac{y_{t+1}}{x_{t+1} x_t} + \frac{y_{t+2}}{x_{t+2} x_{t+1} x_t} + \dots + \frac{y_{n-1}}{x_{n-1} x_{n-2} \dots x_t} + \frac{y_n}{x_n x_{n-1} \dots x_t} \\
a_n &= a_t (x_{t+1} x_{t+2} \dots x_n) + y_{t+1} (x_n x_{n-1} \dots x_{t+2}) + y_{t+2} (x_n x_{n-1} \dots x_{t+3}) \\
&\quad + \dots + y_{n-2} (x_n x_{n-1}) + y_{n-1} x_n + y_n \\
a_n &= a_t \prod_{i=t+1}^n x_i + \sum_{i=t+1}^n \left[y_i \prod_{j=i+1}^n x_j \right]
\end{aligned}$$

6 Problem 2.43.

Solve the recurrence

$$a_n = \frac{\alpha a_{n-1} + \beta}{\gamma a_{n-1} + \delta}$$

for $n > 0$ with $a_0 = 1$.

Solution. First, we guess the form of a_n by getting a_1, a_2 and a_3 .

$$\begin{aligned}
a_0 &= 1 \\
a_1 &= \frac{\alpha a_0 + \beta}{\gamma a_0 + \delta} = \frac{\alpha + \beta}{\gamma + \delta} \\
a_2 &= \frac{\alpha a_1 + \beta}{\gamma a_1 + \delta} = \frac{\alpha \left(\frac{\alpha + \beta}{\gamma + \delta} \right) + \beta}{\gamma \left(\frac{\alpha + \beta}{\gamma + \delta} \right) + \delta} = \frac{\alpha(\alpha + \beta) + \beta(\gamma + \delta)}{\gamma(\alpha + \beta) + \delta(\gamma + \delta)} \\
a_3 &= \frac{\alpha a_2 + \beta}{\gamma a_2 + \delta} = \frac{\alpha \left(\frac{\alpha(\alpha + \beta) + \beta(\gamma + \delta)}{\gamma(\alpha + \beta) + \delta(\gamma + \delta)} \right) + \beta}{\gamma \left(\frac{\alpha(\alpha + \beta) + \beta(\gamma + \delta)}{\gamma(\alpha + \beta) + \delta(\gamma + \delta)} \right) + \delta} \\
&= \frac{\alpha [\alpha(\alpha + \beta) + \beta(\gamma + \delta)] + \beta [\gamma(\alpha + \beta) + \delta(\gamma + \delta)]}{\gamma [\alpha(\alpha + \beta) + \beta(\gamma + \delta)] + \delta [\gamma(\alpha + \beta) + \delta(\gamma + \delta)]}
\end{aligned}$$

Observing the pattern, we have the following system of linear homoge-

neous recurrence relations:

$$\begin{aligned}
a_n &= \frac{b_n}{c_n} \\
b_n &= \alpha b_{n-1} + \beta c_{n-1} \\
c_n &= \gamma b_{n-1} + \delta c_{n-1} \\
b_0 &= 1 \\
c_0 &= 1
\end{aligned}$$

Solving for b_n and c_n , we have

$$\begin{aligned}
\begin{bmatrix} b_n \\ c_n \end{bmatrix} &= \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix} \begin{bmatrix} b_{n-1} \\ c_{n-1} \end{bmatrix} \\
\begin{bmatrix} b_n \\ c_n \end{bmatrix} &= \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}^2 \begin{bmatrix} b_{n-2} \\ c_{n-2} \end{bmatrix} \\
\begin{bmatrix} b_n \\ c_n \end{bmatrix} &= \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}^n \begin{bmatrix} b_0 \\ c_0 \end{bmatrix} = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}^n \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \mathbf{A}^n \begin{bmatrix} 1 \\ 1 \end{bmatrix}
\end{aligned}$$

To solve for the matrix power, we have $\mathbf{A}^n = P D^n P^{-1}$ where P is the matrix of eigenvectors (as columns) of A and D is the diagonal matrix of eigenvalues. $\mathbf{A}$ has the following characteristic polynomial and corresponding eigensystem

$$\begin{aligned}
\lambda^2 - (\alpha + \delta)\lambda + \alpha\delta - \beta\gamma &= 0 \\
\lambda_{1,2} &= \frac{1}{2}(\alpha + \delta \pm \Delta) \\
v_{1,2} &= \left[-\frac{1}{2\gamma}(-\alpha + \delta \mp \Delta), 1 \right]^T \\
\Delta &= \sqrt{(\alpha - \delta)^2 + 4\beta\gamma}
\end{aligned}$$

So the matrix power of $\mathbf{A}$ is

$$\begin{aligned}\mathbf{A}^n &= PD^nP^{-1} \\ \mathbf{A}^n &= \begin{bmatrix} -\frac{1}{2\gamma}(-\alpha + \delta - \Delta) & -\frac{1}{2\gamma}(-\alpha + \delta + \Delta) \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2}(\alpha + \delta + \Delta) & 0 \\ 0 & \frac{1}{2}(\alpha + \delta - \Delta) \end{bmatrix}^n \begin{bmatrix} -\frac{\gamma}{\Delta} & \frac{\alpha - \delta + \Delta}{2\Delta} \\ \frac{\gamma}{\Delta} & \frac{-\alpha + \delta - \Delta}{2\Delta} \end{bmatrix} \\ \mathbf{A}^n &= \begin{bmatrix} \frac{2^{-n-1}((\Delta - \alpha + \delta)(-\Delta + \alpha + \delta)^n + (\Delta + \alpha - \delta)(\Delta + \alpha + \delta)^n)}{\gamma 2^{-n}((\Delta + \alpha + \delta)^n - (-\Delta + \alpha + \delta)^n)} & \frac{\beta 2^{-n}((\Delta + \alpha + \delta)^n - (-\Delta + \alpha + \delta)^n)}{\frac{\Delta}{\Delta}} \\ \frac{\Delta}{\Delta} & \frac{2^{-n-1}((\Delta + \alpha - \delta)(-\Delta + \alpha + \delta)^n + (\Delta - \alpha + \delta)(\Delta + \alpha + \delta)^n)}{\Delta} \end{bmatrix} \\ \mathbf{A}^n \begin{bmatrix} 1 \\ 1 \end{bmatrix} &= \begin{bmatrix} \frac{((-\alpha - 2\beta + \delta)(-\Delta + \alpha + \delta)^n + \Delta(-\Delta + \alpha + \delta)^n + (\alpha + 2\beta - \delta)(\Delta + \alpha + \delta)^n + \Delta(\Delta + \alpha + \delta)^n)}{\frac{\Delta 2^{n+1}}{\Delta 2^{n+1}}} \\ \frac{((\alpha - 2\gamma - \delta)(-\Delta + \alpha + \delta)^n + \Delta(-\Delta + \alpha + \delta)^n + (-\alpha + 2\gamma + \delta)(\Delta + \alpha + \delta)^n + \Delta(\Delta + \alpha + \delta)^n)}{\Delta 2^{n+1}} \end{bmatrix} = \begin{bmatrix} b_n \\ c_n \end{bmatrix}\end{aligned}$$

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Therefore, the solution is

$$a_n = \frac{(-\alpha - 2\beta + \delta)(-\Delta + \alpha + \delta)^n + \Delta(-\Delta + \alpha + \delta)^n + (\alpha + 2\beta - \delta)(\Delta + \alpha + \delta)^n + \Delta(\Delta + \alpha + \delta)^n}{(\alpha - 2\gamma - \delta)(-\Delta + \alpha + \delta)^n + \Delta(-\Delta + \alpha + \delta)^n + (-\alpha + 2\gamma + \delta)(\Delta + \alpha + \delta)^n + \Delta(\Delta + \alpha + \delta)^n}$$

where $\Delta = \sqrt{(\alpha - \delta)^2 + 4\beta\gamma}$.